

RateAR Image & Video Annotating Task

Instructions

Quality Features

Three quality features of interest:

- Placement (sometimes referred to as “floating”)
- Size
- Shadow

All features are rated on a 5-point Likert scale

- 1 translates to worst quality; 5 corresponds to best quality

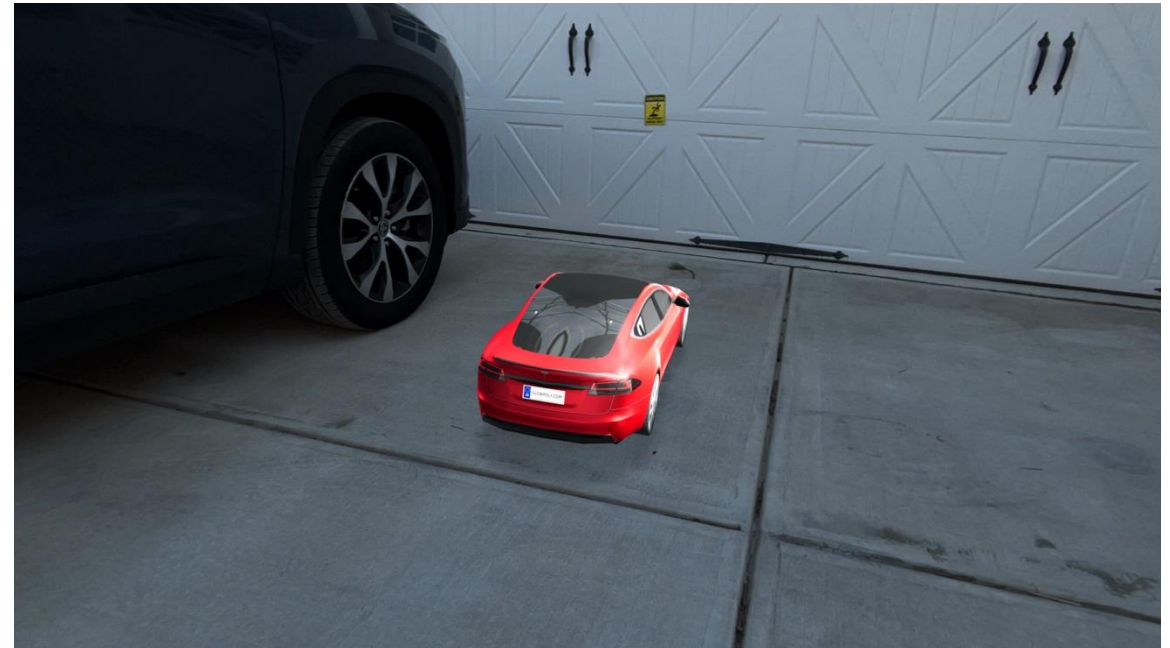
IMPORTANTLY: The quality features are dependent on object/scene context. They are labeled independently of one another!

Instructions: Placement (“Floating”)

Please assess whether the virtual object appears to be floating or placed in a way that doesn't match its surroundings. For example, a basketball can be in the air if it's being thrown and a balloon can float as this is its expected behavior. However, a chair (usually) shouldn't float unless the scene clearly shows furniture designed to levitate.



Quality score: 1
Very apparently floating in the air



Quality score: 5
Appears entirely on the ground

Instructions: Placement (“Floating”) ~ Nuances



Quality score: 2/3

Object appears to be floating, but
hard to determine by how much



Quality score: 3/4

Object doesn't appear to touch the ground
surface, but the shadow's perception could
be manipulating judgment

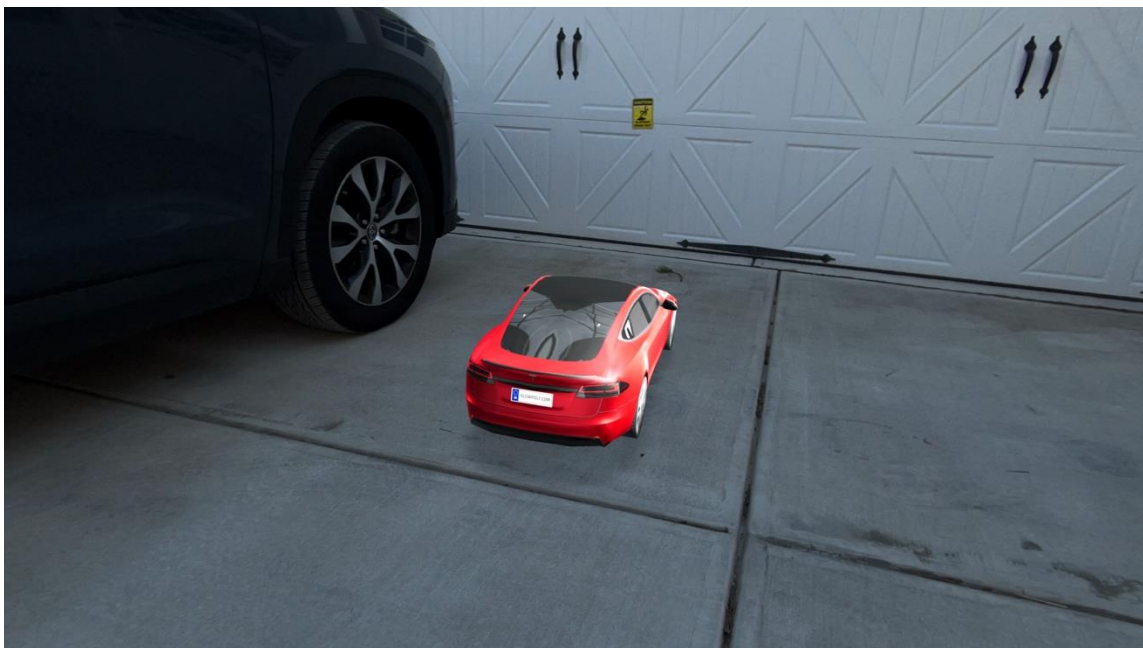
Instructions: Placement (“Floating”) ~ Nuances

If floating behavior is appropriate, assigning a maximum (or near maximum) placement score is the recommended action



Instructions: Size

Please assess whether the virtual object's size appears appropriate for its surroundings. For example, a basketball should be smaller than a person's torso, while a chair shouldn't be as large as an entire room unless the scene specifically calls for exaggerated proportions.

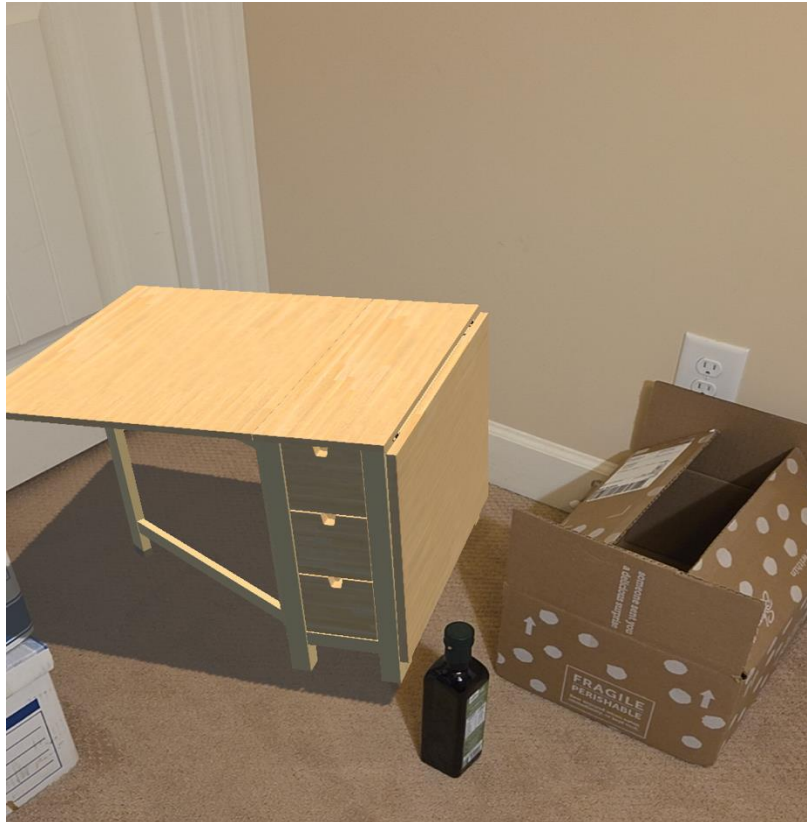


Quality score: 1
Too small



Quality score: 5
(Near) Expected size

Instructions: Size ~ Nuances



Quality score: 2/3

The object is definitively too small, but the context makes it difficult to measure *by how much*



Quality score: 3/4

Approaching a realistic size, given the scene context, but the slight imperfection impacts immersive integration

Instructions: Shadow

Please assess whether the virtual object's shadow appears appropriate for its surroundings, considering natural light sources and the shadows of real objects. For example, the direction of a virtual object's shadow should match that of the surrounding objects. Also consider shadow shape and shade (how light or dark the shadow is), to lesser extents.



Quality score: 1
NO shadow



Quality score: 5
Shadow matches direction of real-world
object shadows, comparable shadow

Instructions: Shadow ~ Nuances

Quality score: ~3

Shadow direction is partially in agreement with real-world object shadows. Virtual content shadow is VERY recognizable from natural shadows, especially due to clipping



Instructions: Shadow ~ Nuances

For some contexts, like medical anatomy content, shadows are not necessary nor expected. In such cases, a shadow score of 5 is appropriate (to indicate that no shadow is CORRECT)!

